

Problem 1. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = e^x + 5$. Is f injective? What is the image of f ? Is f surjective?

Solution. Injectivity: The function f is injective. Let $a, b \in \mathbb{R}$ and suppose that $f(a) = f(b)$. Then $e^a + 5 = e^b + 5$, and subtracting 5 from both sides, we have that $e^a = e^b$. Taking the natural log of both sides gives us that $a = b$.

Image/Surjectivity: The image f is the set $(5, \infty)$. Let $y \in f(\mathbb{R})$. Then $y = e^x + 5$ for some $x \in \mathbb{R}$. Since $e^x > 0$ for all $x \in \mathbb{R}$, $e^x + 5$ is a real number greater than 5, so $y \in (5, \infty)$. Now, let $y \in (5, \infty)$. We want to show there is some $x \in \mathbb{R}$ such that $e^x + 5 = y$. Set $x = \ln(y - 5)$. Since y is a real number greater than 5, $\ln(y - 5) \in \mathbb{R}$, and $f(x) = f(\ln(y - 5)) = e^{\ln(y - 5)} + 5 = y - 5 + 5 = y$. Therefore, $y \in f(\mathbb{R})$, so $f(\mathbb{R}) = (5, \infty)$. Since $(5, \infty) \neq \mathbb{R}$, f is not surjective. $\square$

Problem 2. Let $A \subseteq X$ and $B \subseteq X$ be sets. Prove that $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

Solution. ($\subseteq$): Let $y \in X \setminus (A \cap B)$. We will show that $y \in (X \setminus A) \cup (X \setminus B)$. Since $y \in X \setminus (A \cap B)$, $y \in X$, but $y \notin A \cap B$. Therefore, y is not in A or y is not in B . In other words, $y \in (X \setminus A)$ or $y \in (X \setminus B)$, so $y \in (X \setminus A) \cup (X \setminus B)$.

($\supseteq$): Let $y \in (X \setminus A) \cup (X \setminus B)$. Then $y \in (X \setminus A)$ or $y \in (X \setminus B)$, so $y \notin A$ or $y \notin B$, hence $y \notin A \cap B$. Therefore, $y \in X \setminus (A \cap B)$. $\square$

Problem 3. Let A be the set of polynomials with coefficients in $\mathbb{R}$, i.e.,

$$A = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 : a_i \in \mathbb{R}, n \in \mathbb{N}\}.$$

Let $f: A \rightarrow A$ be the function which sends a polynomial to its derivative. Is f injective? What is the image of f ? Is f surjective?

Solution. Injectivity: The function f is not injective. For example, $x^2 + 2$ and $x^2 + 3$ are two different polynomials in A , but the derivative of both of them is $2x$.

Image/Surjectivity: We will show that the function f is surjective, which implies that $f(A) = A$. Let $g = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in A$. Then the antiderivative of g , $\frac{a_n}{n+1} x^{n+1} + \frac{a_{n-1}}{n} x^n + \cdots + \frac{a_1}{2} x^2 + a_0 x$ has derivative g , so f is surjective. $\square$

Problem 4. Let $f: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ be a function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$, and $f(0) = 1$. Prove that there exists some positive number $a \in \mathbb{R}$ such that $f(n) = a^n$ for all $n \in \mathbb{N}$, $n \geq 1$. Use this to prove that for all $x \in \mathbb{Z}$, $f(x) = a^x$.

Solution. We prove this by induction on $n \in \mathbb{N}$, $n \geq 1$.

Base case: The base case is when $n = 1$. Since $f: \mathbb{R} \rightarrow \mathbb{R}_{>0}$, we have that $f(1) = a$, where a is a positive real number. Therefore, $f(1) = a = a^1$, so the base case is true.

Induction step: Suppose that the claim is true for n . That is, suppose that $f(n) = a^n$. Then $f(n+1) = f(n)f(1) = a^n \cdot a = a^{n+1}$.

Now, we show that for any integer $x \in \mathbb{Z}$, $f(x) = a^x$. If $x = 0$, we are given that $f(0) = 1 = a^0$. If $x > 0$, then $x \in \mathbb{N}$, $x \geq 1$, so $f(x) = a^x$ from above. If $x < 0$, then $-x = n$ for some $n \in \mathbb{N}$, $n \geq 1$. We have

$$f(x-x) = f(0) = 1 = f(x)f(-x).$$

Hence $f(x) = 1/f(-x) = 1/a^{-x} = a^x$.

□